NLO Gluon Production: BFKL, JIMWLK, Running-Coupling, DGLAP and Finite Terms

A step-by-step derivation of every term from the NLO single-gluon spectrum

$d_\perp = 2-\epsilon$ (so $2/\epsilon = 1/\hat\epsilon$ for $D = 4-2\hat\epsilon$), $\overline{\rm MS}$ scale $\mu \equiv \mu_{\overline{MS}}$, $c_0 \equiv 2e^{-\gamma_E}$, $\int_x \equiv \int d^2x$, pure glue ($\beta_0 = \frac{11}{3}N_c$).

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0. Starting expression

$$\frac{d^3N_{\rm NLO}}{d^3k} = \underbrace{\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4}-1\Big]\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)/\xi}\Big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}}_{\text{line (I)}}$$

$$+\underbrace{\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\Big[\log\frac{k^+}{\Lambda}+\log\Big(\frac{\vee}{k^+}-1\Big)\Big]\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[\cdots\Big]\mathcal{O}}_{\text{line (II)}}$$

$$+\underbrace{\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^{k^+/(k^+\!+\Lambda)}d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)/\xi}\Big[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-d_\perp\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\Big]\mathcal{O}}_{\text{line (III)}}$$

Here $[\cdots]$ in line (II) is the same bracket as in line (I), and $\mathcal{O} = [U^{e'c}(y') - U^{e'c}(x')][U^{ec}(y) - U^{ec}(x)]\,\rho^{e'}(x')\rho^{e}(x)$.

> The last Wilson line is written $U^{ce}(x)$ in the original. It is taken to be $U^{ec}(x)$ throughout; it multiplies every term identically, so this does not affect the separation.

1. Notation

$$C \equiv \frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}, \qquad a \equiv \frac{k^+}{\vee}, \qquad b \equiv \frac{k^+}{k^+\!+\Lambda}, \qquad r \equiv y'-y,$$

$$\mathcal{A} \equiv \frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}, \qquad \mathcal{B} \equiv \frac{(x-y)\cdot r\,(x'-y')\cdot r}{(x-y)^2(x'-y')^2\,r^2}, \qquad K_{ww'}(z) \equiv \frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2},$$

$$\Phi_W(q) \equiv \int_{x,y,x',y'} e^{-iq\cdot r}\,W\,\mathcal{O}.$$

All three lines carry the same prefactor C (check: N_c simply moves between factors). $\Phi_{\mathcal{A}}(q)$ is the LO structure at transverse momentum q:

$$\Phi_{\mathcal{A}}(q) = a_i(q)\,a_i^\dagger(q), \qquad a_i(q) = \int_{x,y} e^{iq\cdot y}\frac{(x-y)_i}{(x-y)^2}\big[U(y)-U(x)\big]\rho(x).$$

In this notation the expression is

$$\boxed{\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} &= C\Big\{\int_a^1\frac{d\xi}{\xi^2}\,p(\xi)\,\Phi_{\mathcal{A}L}\big(\tfrac{k}{\xi}\big) + \Big[\log\frac{k^+}{\Lambda}+\log\Big(\frac{\vee}{k^+}-1\Big)\Big]\Phi_{\mathcal{A}L}(k) + \int_a^b d\xi\,\frac{1-\xi}{\xi}\,\Phi_{\mathcal{A}-d_\perp\mathcal{B}}\big(\tfrac{k}{\xi}\big)\Big\},\\ p(\xi) &\equiv \xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+},\end{aligned}}$$

where L is the bracket, treated in Step 1.

2. Step 1: expand the bracket L

Pull out $d_\perp/2$ and combine γ_E with the log, using $\log\frac{r^2\mu^2e^{2\gamma_E}}{4} = \log\frac{r^2\mu^2}{c_0^2}$:

$$L = -\frac{d_\perp}{\epsilon} - d_\perp\gamma_E - \frac{d_\perp}{2}\log\frac{r^2\mu^2}{4} - 1 = -\frac{d_\perp}{2}\Big[\frac{2}{\epsilon}+\log\frac{r^2\mu^2}{c_0^2}\Big] - 1.$$

Insert $d_\perp/2 = 1-\epsilon/2$. The $O(\epsilon)\times\frac{2}{\epsilon}$ cross term gives $+1$, which cancels the -1:

$$L = -\frac{2}{\epsilon} - \log\frac{r^2\mu^2}{c_0^2} + 1 - 1 + O(\epsilon) \qquad\Longrightarrow\qquad \boxed{L(r) = -\frac{2}{\epsilon} - \log\frac{r^2\mu^2}{c_0^2} + O(\epsilon)}$$

The remaining $O(\epsilon)$ terms can be dropped: every quantity L multiplies is finite. These are the ξ-integrals at fixed $\vee, \Lambda$, the rapidity logs, and an ϵ-independent $g \to gg$ splitting function.

Meaning of L. With $\int\frac{d^dq}{(2\pi)^d}\frac{e^{iq\cdot r}}{(q^2)^\alpha} = \frac{\Gamma(d/2-\alpha)}{4^\alpha\pi^{d/2}\Gamma(\alpha)}(r^2)^{\alpha-d/2}$ at $\alpha=1$, $d=2-\epsilon$, and the $\overline{\rm MS}$ rescaling $\mu^2 \to \mu^2e^{\gamma_E}/4\pi$,

$$4\pi\mu^\epsilon\int\frac{d^dq}{(2\pi)^d}\frac{e^{iq\cdot r}}{q^2} = \Gamma\big(-\tfrac{\epsilon}{2}\big)\Big(\frac{r^2\mu^2e^{\gamma_E}}{4}\Big)^{\epsilon/2} = -\frac{2}{\epsilon} - \log\frac{r^2\mu^2}{c_0^2} + O(\epsilon) = L(r).$$

In two dimensions, $\int d^2z\,K_{yy'}(z) = \int d^2q\,\frac{e^{iq\cdot r}}{q^2}$. So L is the integral over the position z of the unobserved gluon (with no Wilson line at z), and $1/\epsilon$ is its final-state collinear divergence: relative transverse momentum $q \to 0$, i.e. large separation.

3. Step 2: the blue bracket is a ξ-integral

From the definitions of a and b:

$$\frac{b}{1-b} = \frac{k^+}{\Lambda}, \qquad \frac{a}{1-a} = \frac{k^+}{\vee - k^+}.$$

With $\int\frac{d\xi}{\xi(1-\xi)} = \log\frac{\xi}{1-\xi}$:

$$\int_a^b \frac{d\xi}{\xi(1-\xi)} = \log\frac{b}{1-b} - \log\frac{a}{1-a} = \log\frac{k^+}{\Lambda} + \log\Big(\frac{\vee}{k^+} - 1\Big) \qquad \text{exactly.}$$

So line (II) is $C\,\Phi_{\mathcal{A}L}(k)\int_a^b \frac{d\xi}{\xi(1-\xi)}$: an integrand $\frac{1}{\xi(1-\xi)}$ multiplied by the phase at its $\xi = 1$ value.

4. Step 3: identify the plus prescription and combine lines (I) and (II)

The $\xi/(1-\xi)$ term of line (I), before regularization, is

$$\int_a^b \frac{d\xi}{\xi^2}\,\frac{\xi}{1-\xi}\,e^{-ik\cdot r/\xi} = \int_a^b d\xi\,\frac{e^{-ik\cdot r/\xi}}{\xi(1-\xi)}.$$

Adding and subtracting the integrand at its $\xi = 1$ value $e^{-ik\cdot r}$ gives

$$\int_a^b d\xi\,\frac{e^{-ik\cdot r/\xi}}{\xi(1-\xi)} = \int_a^1 d\xi\,\frac{e^{-ik\cdot r/\xi} - e^{-ik\cdot r}}{\xi(1-\xi)} + e^{-ik\cdot r}\int_a^b \frac{d\xi}{\xi(1-\xi)} + O\Big(\frac{\Lambda}{k^+}\Big).$$

The first integral is regular at $\xi = 1$, so its upper limit b can be replaced by 1; the error is $O(1-b) = O(\Lambda/k^+)$. By Step 2, the second term is exactly line (II). So line (I) + line (II) is the decomposition above, and the plus prescription of line (I) means

$$\int_a^1 \frac{d\xi}{\xi^2}\,\frac{\xi}{(1-\xi)_+}\,e^{-ik\cdot r/\xi} \equiv \int_a^1 d\xi\,\frac{e^{-ik\cdot r/\xi} - e^{-ik\cdot r}}{\xi(1-\xi)}. \qquad \text{(U)}$$

Convert (U) to the standard plus distribution. Let $f(\xi) \equiv e^{-ik\cdot r/\xi}/\xi$, so $f(1) = e^{-ik\cdot r}$. The standard $[0,1]$ distribution restricted to $[a,1]$ is

$$\text{(S)} \equiv \int_a^1 d\xi\,\frac{f(\xi)}{(1-\xi)_+} = \int_a^1 d\xi\,\frac{f(\xi)-f(1)}{1-\xi} + f(1)\log(1-a).$$

Then

$$\text{(U)} = \int_a^1 d\xi\,\frac{f(\xi)-f(1)/\xi}{1-\xi} = \int_a^1 d\xi\,\frac{f(\xi)-f(1)}{1-\xi} + f(1)\int_a^1 d\xi\,\frac{1-1/\xi}{1-\xi} = \int_a^1 d\xi\,\frac{f(\xi)-f(1)}{1-\xi} - f(1)\log\frac{1}{a},$$

since $\frac{1-1/\xi}{1-\xi} = -\frac{1}{\xi}$. Therefore

$$\text{(U)} = \text{(S)} - f(1)\Big[\log(1-a) + \log\frac{1}{a}\Big] = \text{(S)} - e^{-ik\cdot r}\log\Big(\frac{\vee}{k^+} - 1\Big).$$

Combine. Insert this into line (I) and add line (II). The $\log(\vee/k^+ - 1)$ cancels exactly:

$$\text{(I)} + \text{(II)} = C\Big\{\int_a^1 \frac{d\xi}{\xi^2}\,p_{\rm S}(\xi)\,\Phi_{\mathcal{A}L}\big(\tfrac{k}{\xi}\big) + \log\frac{k^+}{\Lambda}\,\Phi_{\mathcal{A}L}(k)\Big\},$$

where $p_{\rm S}$ is p with the standard plus distribution. The explicit $\log(\vee/k^+ - 1)$ of line (II) is the price of the subtraction (U); it is not a separate rapidity log. All $\log(\vee/k^+)$ dependence now sits in the lower limit of the $(1-\xi)/\xi$ term (Step 5).

> This identification follows from the exact identity of Step 2. If your $(1-\xi)_+$ was instead already the standard one, the $\log(\vee/k^+ - 1)$ of line (II) would not cancel, and it would double count the BFKL term of Step 5.

5. Step 4: complete the splitting function and isolate the running coupling

The full pure-glue $g \to g$ splitting function is

$$P_{gg}(\xi) = 2N_c\Big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\Big] + \frac{\beta_0}{2}\,\delta(1-\xi), \qquad \beta_0 = \frac{11}{3}N_c.$$

The momentum sum rule fixes the δ-coefficient:

$$\int_0^1 d\xi\,\xi\,p_{\rm S}(\xi) = \int_0^1 \frac{\xi^2-1}{1-\xi}\,d\xi + \int_0^1 (1-\xi)\,d\xi + \int_0^1 \xi^2(1-\xi)\,d\xi = -\frac{3}{2} + \frac{1}{2} + \frac{1}{12} = -\frac{11}{12},$$

$$\int_0^1 d\xi\,\xi\,P_{gg} = 2N_c\Big(-\frac{11}{12}\Big) + \frac{11N_c}{6} = 0.$$

Hence $p_{\rm S} = \frac{P_{gg}}{2N_c} - \frac{11}{12}\delta(1-\xi)$. The δ-term evaluates at $\xi = 1$ (with $\int_a^1 \delta(1-\xi)\,d\xi = 1$), so

$$\int_a^1 \frac{d\xi}{\xi^2}\,p_{\rm S}\,\Phi_{\mathcal{A}L}\big(\tfrac{k}{\xi}\big) = \int_a^1 \frac{d\xi}{\xi^2}\,\frac{P_{gg}(\xi)}{2N_c}\,\Phi_{\mathcal{A}L}\big(\tfrac{k}{\xi}\big) - \frac{11}{12}\,\Phi_{\mathcal{A}L}(k).$$

The last term is the **running-coupling term** (Section 10). It carries the β_0 coefficient that the plus distribution did not produce.

6. Step 5: the $\log(\vee/k^+)$ hidden in the P_{gg} convolution, which is the BFKL term

6.1 Only the $(1-\xi)/\xi$ term and the $-\log r^2$ part contribute

A $\log(1/a) = \log(\vee/k^+)$ can only come from $\xi \to a \to 0$, where Φ is probed at $k/\xi \to \infty$. Split

$$L = L_0 - \log r^2, \qquad L_0 \equiv -\frac{2}{\epsilon} - \log\frac{\mu^2}{c_0^2} \quad (\text{no } r).$$

- **The L_0 part.** It multiplies $\Phi_{\mathcal{A}}(k/\xi)$. As $y \to x$, $[U(y) - U(x)] \simeq (y-x)_j\partial_j U$ cancels the $1/|x-y|$ of the kernel. The remaining bounded function has $a_i \sim q^{-2}$, so $\Phi_{\mathcal{A}}(q) \sim q^{-4}$ (the LO $1/k^4$ tail). Even the most singular weight $\frac{1-\xi}{\xi^3}$ then gives $\sim \xi$: no log. So the pole and $\log\mu^2$ do not enter the BFKL term.
- **The $-\log r^2$ part.** Section 6.2 shows $\xi^{-2}\Phi_{\mathcal{A}(-\log r^2)}(k/\xi) \to \mathcal{S}$, a constant. Multiplied by the weights:
  - $\frac{1}{\xi^2}\,\xi(1-\xi)\cdot\xi^2\mathcal{S}$ goes like ξ: no log.

- $\frac{1}{\xi^2}\,\frac{\xi}{(1-\xi)_+}\cdot\xi^2\mathcal{S}$ goes like ξ: no log. The plus subtraction acts at $\xi=1$ only.
- $\frac{1}{\xi^2}\,\frac{1-\xi}{\xi}\cdot\xi^2\mathcal{S}\to\mathcal{S}/\xi$: this gives $\log(1/a)$.

6.2 The limit $\mathcal{S}$

(a) Rescale the dipole. At fixed x,x',y, put $y'=y+\xi s$:

$$d^2y'=\xi^2 d^2s,\qquad \frac{k\cdot(y'-y)}{\xi}=k\cdot s,\qquad -\log(y'-y)^2=-\log s^2-\log\xi^2.$$

This gives

$$\xi^{-2}\,\Phi_{\mathcal{A}(-\log r^2)}\big(\tfrac{k}{\xi}\big)=\int_{x,y,x',s} e^{-ik\cdot s}\big[-\log s^2-\log\xi^2\big]\,\mathcal{A}(x,y,x',y+\xi s)\,\mathcal{O}(y,y+\xi s).$$

The $-\log\xi^2$ piece is $-\log\xi^2\,\xi^{-2}\Phi_{\mathcal{A}}(k/\xi)\sim\xi^2\log\xi\to 0$.

(b) Take $\xi\to 0$. For fixed x,x',y, $\mathcal{A}\mathcal{O}$ is smooth in y' at $y'=y$, so the limit is reached with $O(\xi)$ corrections. Rename $y\to z$; the hard parent then sits at a single point z in the amplitude and the conjugate:

$$\mathcal{S}=J(k)\int_{x,x',z}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\big[U^{e'c}(z)-U^{e'c}(x')\big]\big[U^{ec}(z)-U^{ec}(x)\big]\rho^{e'}(x')\rho^{e}(x),$$

$$J(k)\equiv\int d^2s\,e^{-ik\cdot s}\big(-\log s^2\big).$$

(c) Fourier transform of the log. From $\nabla^2\log s^2=4\pi\delta^2(s)$, multiplying by $e^{-ik\cdot s}$ and integrating by parts twice:

$$-k^2\int d^2s\,e^{-ik\cdot s}\log s^2=4\pi\quad\Longrightarrow\quad J(k)=\frac{4\pi}{k^2}\qquad(k\neq 0).$$

Any constant added to $\log s^2$ only gives $\delta^2(k)$, which vanishes for $k\neq 0$.

(d) Write it as the soft-gluon kernel. Using $u_i/u^2=\frac{1}{2}\partial_i\log u^2$:

$$I_i(k)\equiv\int d^2u\,e^{ik\cdot u}\frac{u_i}{u^2}=-\frac{ik_i}{2}\int d^2u\,e^{ik\cdot u}\log u^2=2\pi i\frac{k_i}{k^2},\qquad I_i(-k)=-I_i(k).$$

With $u=w-z$ and $u'=w'-z$, for any z:

$$\int_{w,w'}e^{-ik\cdot(w'-w)}K_{ww'}(z)=I_i(k)\,I_i(-k)=\frac{4\pi^2}{k^2}\qquad\Longrightarrow\qquad J(k)=\frac{1}{\pi}\int_{w,w'}e^{-ik\cdot(w'-w)}K_{ww'}(z).$$

$$\boxed{\mathcal{S}=\frac{1}{\pi}\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\big[U^{e'c}(z)-U^{e'c}(x')\big]\big[U^{ec}(z)-U^{ec}(x)\big]\rho^{e'}(x')\rho^{e}(x)}$$

6.3 Exact subtraction

Since $\int_a^1 d\xi/\xi=\log(\vee/k^+)$ exactly, write

$$\int_a^1\frac{d\xi}{\xi^2}\,\frac{P_{gg}}{2N_c}\,\Phi_{\mathcal{A}L}\big(\tfrac{k}{\xi}\big)=\mathcal{S}\log\frac{\vee}{k^+}+\int_a^1 d\xi\Big[\frac{1}{\xi^2}\frac{P_{gg}}{2N_c}\Phi_{\mathcal{A}L}\big(\tfrac{k}{\xi}\big)-\frac{\mathcal{S}}{\xi}\Big].$$

By Sections 6.1–6.2, the bracket on the right is integrable at $\xi=0$, so its lower limit can be set to 0, up to $O(k^+/\vee)$.

7. Step 6: split the subtracted convolution into DGLAP and finite parts

The convolution probes the parent momentum $q=k/\xi$. Split L at that scale:

$$L=\underbrace{\Big[-\frac{2}{\epsilon}-\log\frac{\xi^2\mu^2}{k^2}\Big]}_{\text{pole}+\log(\mu^2/q^2)}\underbrace{-\log\frac{k^2r^2}{\xi^2c_0^2}}_{-\log(q^2r^2/c_0^2)}.$$

The first bracket has no r-dependence, so it multiplies $\Phi_{\mathcal{A}}(k/\xi)$. This is the standard DGLAP structure $-[1/\hat{\epsilon}+\log(\mu^2/q^2)]\,P\otimes\mathrm{LO}$. By Section 6.1 it is integrable at $\xi\to 0$ with no subtraction. The $\mathcal{S}/\xi$ subtraction therefore belongs entirely to the second piece. Only $-\log r^2$ feeds $\mathcal{S}$; the extra $\log(\xi^2/k^2)$ multiplies $\Phi_{\mathcal{A}}$ and vanishes as $\xi^2\log\xi$.

8. Step 7: line (III) is finite

- $d_\perp\to 2$. Line (III) has no pole, so $(d_\perp-2)\mathcal{B}=-\epsilon\,\mathcal{B}\to 0$.
- **Upper limit, no** $\log(k^+/\Lambda)$. The weight $(1-\xi)/\xi$ vanishes at $\xi=1$, so $\int_b^1(1-\xi)\,d\xi=O(\Lambda^2/k^{+2})$ and $b\to 1$.
- **Lower limit, no** $\log(\vee/k^+)$. As $r\to 0$, $(\mathcal{A}-2\mathcal{B})\,\mathcal{O}\to\frac{(x-y)_i(x'-y)_j}{(x-y)^2(x'-y)^2}(\delta_{ij}-2\hat{r}_i\hat{r}_j)\,\mathcal{O}(y,y)$. The matrix $\delta_{ij}-2\hat{r}_i\hat{r}_j=-\begin{pmatrix}\cos 2\phi & \sin 2\phi\\ \sin 2\phi & -\cos 2\phi\end{pmatrix}$ is traceless and has angular average 0. Using $\int d\phi\,e^{\pm 2i\phi}e^{-iqr\cos(\phi-\phi_q)}=-2\pi J_2(qr)\,e^{\pm 2i\phi_q}$ and $\int_0^\infty r\,J_2(qr)\,dr=2/q^2$:

  $$\int d^2r\,e^{-iq\cdot r}\big(\delta_{ij}-2\hat{r}_i\hat{r}_j\big)=-\frac{4\pi}{q^2}\big(\delta_{ij}-2\hat{q}_i\hat{q}_j\big).$$

  So $\Phi_{\mathcal{A}-2\mathcal{B}}(k/\xi)\sim\xi^2$, and $\frac{1-\xi}{\xi}\,\xi^2\sim\xi$: no log. The limit $a\to 0$ costs $O(k^{+2}/\vee^2)$.

9. Result: all terms

Collecting Steps 1–7, with $\mathcal{S}$ from Section 6.2:

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{d^3N^{\rm BFKL}}{d^3k} + \frac{d^3N^{\rm JIMWLK}}{d^3k} + \frac{d^3N^{\rm RC}}{d^3k} + \frac{d^3N^{\rm DGLAP}}{d^3k} + \frac{d^3N^{\rm finite}}{d^3k} + O\Big(\frac{k^+}{\vee}, \frac{\Lambda}{k^+}, \epsilon\Big).$$

9.1 BFKL, $\propto \log(\vee/k^+)$

$$\frac{d^3N^{\rm BFKL}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2} \times \big[U^{e'c}(z) - U^{e'c}(x')\big]\big[U^{ec}(z) - U^{ec}(x)\big]\rho^{e'}(x')\rho^{e}(x)$$

This is $C\,\mathcal{S}\log(\vee/k^+)$; the prefactor $C/\pi = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}$ includes the $1/\pi$ of $\mathcal{S}$.

9.2 JIMWLK, $\propto \log(k^+/\Lambda)$

$$\frac{d^3N^{\rm JIMWLK}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[-\frac{2}{\epsilon} - \log\frac{(y'-y)^2\mu^2}{c_0^2}\Big]\mathcal{O}$$

Using $L = \frac{1}{\pi}\int_z K_{yy'}(z)$ (dimensionally regularized z-integral, Step 1), this is

$$\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)}K_{yy'}(z)\,\mathcal{A}\,\mathcal{O}.$$

It has the same prefactor as BFKL. The roles are reversed: here the soft, unobserved gluon z is emitted by the measured gluon (y,y'); in BFKL the soft measured gluon (w,w') is emitted by the hard gluon z.

9.3 Running coupling

$$\frac{d^3N^{\rm RC}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^2}{4\pi^2}\frac{g^2}{4\pi^2}\frac{1}{k^+}\,\frac{11N_c}{6}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[\frac{2}{\epsilon} + \log\frac{(y'-y)^2\mu^2}{c_0^2}\Big]\mathcal{O}$$

This is $-\frac{11}{12}C\,\Phi_{\mathcal{A}L}(k)$, with $\frac{11N_c}{6} = \frac{\beta_0}{2}$; in terms of α_s it is $\frac{\alpha_s\beta_0}{2\pi}\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{1}{k^+}\big[\frac{2}{\epsilon}\Phi_{\mathcal{A}} + \Phi_{\mathcal{A}\log(r^2\mu^2/c_0^2)}\big](k)$.

- In pure dimensional regularization the on-shell self-energy is scaleless ($1/\epsilon_{\rm UV} - 1/\epsilon_{\rm IR} = 0$). So the pole can be read as the UV pole removed by $\overline{\rm MS}$ charge renormalization of the LO.
- The log then makes the coupling run to the dipole scale: $\alpha_s(\mu^2) \to \alpha_s(c_0^2/r^2) = \alpha_s(\mu^2)\big[1 + \frac{\alpha_s\beta_0}{4\pi}\log\frac{r^2\mu^2}{c_0^2}\big]$.
- With quarks, $\beta_0 \to \frac{11N_c - 2n_f}{3}$, and the $g \to q\bar{q}$ real channel must also be added.

9.4 DGLAP

$$\frac{d^3N^{\rm DGLAP}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^2}{4\pi^2}\frac{g^2}{4\pi^2}\frac{1}{k^+}\int_0^1\frac{d\xi}{\xi^2}\,P_{gg}(\xi)\Big[-\frac{2}{\epsilon} - \log\frac{\xi^2\mu^2}{k^2}\Big]\int_{x,y,x',y'} e^{-ik\cdot(y'-y)/\xi}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O}$$

Here P_{gg} is the full function of Step 4 with the standard plus distribution, and $C/(2N_c)$ has been used. This is the timelike $g \to g$ fragmentation convolution: $\frac{d\xi}{\xi^3} \times \frac{dN_{\rm LO}}{d^3p}$ with $\frac{dN_{\rm LO}}{d^3p} \propto \frac{1}{p^+}$, at the parent momentum $p^+ = k^+/\xi$, $p = k/\xi$. The $-2/\epsilon = -1/\hat{\epsilon}$ is removed by the $\overline{\rm MS}$ counterterm of $D_g(\xi,\mu^2)$. The $\log(\xi^2\mu^2/k^2) = \log(\mu^2/p^2)$ is the factorization-scale log, natural choice $\mu = |k|/\xi$.

9.5 Finite terms

$$\frac{d^3N^{\rm finite}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\Bigg\{\int_0^1 d\xi\Big[\frac{1}{\xi^2}\frac{P_{gg}(\xi)}{2N_c}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)/\xi}\,\mathcal{A}\Big(-\log\frac{k^2(y'-y)^2}{\xi^2c_0^2}\Big)\mathcal{O} - \frac{\mathcal{S}}{\xi}\Big] + \int_0^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)/\xi}\Big[\mathcal{A} - 2\mathcal{B}\Big]\mathcal{O}\Bigg\}$$

- **First integral.** This is the non-log remainder of the P_{gg} convolution. The $-\mathcal{S}/\xi$ subtraction is exactly what was moved into BFKL, so the integrand is regular at $\xi = 0$.
- **Second integral.** This is line (III) at $d_\perp = 2$, with the limits extended to $0, 1$.
- Both are free of poles, of $\log\mu$, and of rapidity logs.

10. Bookkeeping: the pieces add back up

Step	Identity
3	(I) + (II) $= C\{\int_a^1 \frac{d\xi}{\xi^2} p_{\rm S}\Phi_{\mathcal{A}L}(k/\xi) + \log\frac{k^+}{\Lambda}\Phi_{\mathcal{A}L}(k)\}$; the second term is JIMWLK
4	$C\int \frac{d\xi}{\xi^2} p_{\rm S}\Phi_{\mathcal{A}L} = C\int \frac{d\xi}{\xi^2}\frac{P_{gg}}{2N_c}\Phi_{\mathcal{A}L} + \text{RC}$
5	$C\int_a^1 \frac{d\xi}{\xi^2}\frac{P_{gg}}{2N_c}\Phi_{\mathcal{A}L} = \text{BFKL} + C\int_0^1 d\xi[\cdots - \mathcal{S}/\xi]$
6	$C\int_0^1 d\xi[\cdots - \mathcal{S}/\xi] = \text{DGLAP} + \text{first finite integral, since} - \log\frac{\xi^2\mu^2}{k^2} - \log\frac{k^2r^2}{\xi^2 c_0^2} = -\log\frac{r^2\mu^2}{c_0^2}$
7	(III) $=$ second finite integral

The sum reproduces the starting expression exactly, up to $O(k^+/\vee)$, $O(\Lambda/k^+)$ and $O(\epsilon)$. The RC term and the $\delta(1-\xi)$ parts of DGLAP and of the first finite integral cancel identically: the δ-term was added and subtracted.

11. Verification of the finite terms

11.1 First finite integral (remainder of the P_{gg} convolution)

(a) It recombines with DGLAP. The two logs add to the original one:

$$-\log\frac{\xi^2\mu^2}{k^2} - \log\frac{k^2r^2}{\xi^2 c_0^2} = -\log\frac{r^2\mu^2}{c_0^2}.$$

So DGLAP + first finite integral $= C\int_0^1 d\xi\big[\xi^{-2}\frac{P_{gg}}{2N_c}\Phi_{\mathcal{A}L}(k/\xi) - \mathcal{S}/\xi\big]$, which is Step 5. ✓

(b) A cleaner form. Rescale $y' = y + \xi s$ as in Section 6.2. The finite log is then ξ-independent,

$$-\log\frac{k^2r^2}{\xi^2 c_0^2}\Big|_{r=\xi s} = -\log\frac{k^2 s^2}{c_0^2},$$

and the first finite integral becomes

$$C\int_0^1 d\xi\Big[\frac{P_{gg}(\xi)}{2N_c}\,H(\xi) - \frac{\mathcal{S}}{\xi}\Big], \qquad H(\xi) \equiv \int_{x,y,x',s} e^{-ik\cdot s}\Big(-\log\frac{k^2s^2}{c_0^2}\Big)\mathcal{A}(x,y,x',y+\xi s)\,\mathcal{O}(y,y+\xi s).$$

Here $H(1)$ is the unrescaled $\Phi_{\mathcal{A}(-\log(k^2r^2/c_0^2))}(k)$, and the plus distribution acts on $\xi H(\xi)$ as in the original expression.

(c) It converges at $\xi = 0$. As $\xi \to 0$, $H(\xi) \to H(0)$. The constant $-\log(k^2/c_0^2)$ gives only $\delta^2(k)$, and $\int d^2s\, e^{-ik\cdot s}(-\log s^2) = 4\pi/k^2$, so $H(0) = \mathcal{S}$. The next term is $O(\xi)$: the Fourier transform of $s_i \log s^2$ falls as $|k|^{-3}$. Using $\frac{P_{gg}}{2N_c} = \frac{1}{\xi} - 1 + 2\xi + O(\xi^3)$ for $\xi < 1$:

$$\frac{P_{gg}}{2N_c}H - \frac{\mathcal{S}}{\xi} = \frac{H(\xi)-\mathcal{S}}{\xi} - H(\xi) + O(\xi) = O(1).$$

So the integrand is bounded at $\xi = 0$, and moving the lower limit from a to 0 costs $O(k^+/\vee)$. ✓

(d) Smoothness of $\mathcal{A}\,\mathcal{O}$ at $y' = y$. The only candidate singularities in y' are at $y' = x'$. There the pole of $(x'-y')/(x'-y')^2$ is cancelled by $U(y') - U(x') \simeq (y'-x')\cdot\partial U$, the same mechanism that gives the LO q^{-4} tail. ✓

(e) The $\delta(1-\xi)$ part. It is $\frac{11}{12}C\,H(1)$. Together with the δ-part of DGLAP it gives $\frac{11}{12}C\,\Phi_{\mathcal{A}L}(k)$, which cancels the RC term identically, as it must after the add-and-subtract. ✓

11.2 Second finite integral (line III)

- Its upper limit $b \to 1$ costs $\frac{1}{2}(1-b)^2 = O(\Lambda^2/k^{+2})$, since the weight vanishes at $\xi = 1$. ✓
- Its lower limit $a \to 0$ costs $O(a^2)$, since $\Phi_{\mathcal{A}-2\mathcal{B}}(k/\xi) \sim \xi^2$ (Step 7). ✓
- $d_\perp \to 2$: the difference is $-\epsilon\,\mathcal{B}$ times a finite integral, which vanishes as $\epsilon \to 0$. ✓

11.3 Scheme caveat

The finite terms are $\overline{\rm MS}$ finite terms. They assume that the objects multiplying each $1/\epsilon$ (in DGLAP, RC and JIMWLK) are subtracted as the same $d_\perp$-dimensional objects by the corresponding counterterms or virtual graphs. Then the $O(\epsilon)\times\frac{1}{\epsilon}$ cross terms cancel. If you instead evaluate those transverse integrals in strictly 2 dimensions before subtracting, extra constants (for example $\log 4\pi$ or γ_E pieces) move into the finite part.

12. Remarks and checks

1. **Plus prescription.** The separation of Step 3 is the one forced by the exact identity of Step 2. If your $(1-\xi)_+$ is the standard one, line (II) keeps an extra $C\log(\vee/k^+ - 1)\,\Phi_{\mathcal{A}L}(k)$. At large k that extra term tends to the BFKL term, so it would double count.
2. **The JIMWLK term carries the collinear pole.** It is a soft–collinear double log, $\log(k^+/\Lambda)$ times $1/\epsilon$. In the complete NLO the $\frac{1}{\epsilon}\log\frac{k^+}{\Lambda}$ must cancel against virtual graphs with the same cutoff Λ and against the $U(z)$ terms of the JIMWLK Hamiltonian, which are not in the expression shown. Do not also absorb it into DGLAP. The BFKL term, by contrast, comes out pole-free (Section 6.1).
3. **Normalization.** Relative to $\mathcal{N}\Phi_{\mathcal{A}}$ with $\mathcal{N} = \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{1}{k^+}$, the DGLAP coefficient is $(\alpha_s/\pi)P_{gg}$ and the RC coefficient is $\alpha_s\beta_0/2\pi$. Both are the textbook values, $(\alpha_s/2\pi)P_{gg}$ and $\alpha_s\beta_0/4\pi$, if the LO is $dN_{\rm LO}/d^3k = 2\mathcal{N}\Phi_{\mathcal{A}}(k)$. The two checks are consistent with each other. If the LO is $\mathcal{N}\Phi_{\mathcal{A}}$, look for a stray factor of 2, for example the identical-gluon factor $\frac{1}{2}$.
4. **Assumptions.** Two are used: smoothness of $\mathcal{A}\,\mathcal{O}$ in y' at $y' = y$ for fixed x, x', y (Step 5), and the LO q^{-4} tail. Both are standard; a target field that is rough at short distances would need a separate check.